

IMEN266 Operations Research II — Homework 1 (Ch. 1–3)

Fall 2026

Ungraded, but: **some Part-A problems may reappear on the exams.**

Submit to PLMS: one PDF (parts A, B, D write-ups) + your HW1.ipynb.

AI rules for this homework (details: docs/ai.usage.guide.pdf)

- **Part A:** solve *alone first*, closed-book style, on paper. Only afterwards may you check with an AI — and if the AI changes your mind, record why in Part D.
- **Parts B–C:** AI collaboration is expected. You remain responsible for every claim you submit.
- **Part D:** the prompt log is mandatory whenever AI was used.

Part A — Exam-style warm-up (no AI on first attempt)

- A1. A fab has two lithography tools. Tool 1 processes 60% of all wafers and produces defective wafers with probability 0.01; Tool 2 processes 40% with defect probability 0.03. A randomly selected wafer is found defective. What is the probability it was processed by Tool 2?
- A2. A production line has yield 90% (each part independently defective with probability 0.1).
- (i) What is the probability that the 5th part produced is the 2nd defective one?
 - (ii) What is the expected number of parts produced up to and including the 3rd defective one?
- A3. Let X have pdf $f(x) = cx(2 - x)$ for $0 < x < 2$ (and 0 otherwise). Find c , the cdf F , $E[X]$, and $\text{Var}(X)$.

Part B — Critique an AI solution (find the flaws)

A student asked an AI assistant:

*“A cancer test is 95% accurate on people who have cancer and 96% accurate on people who don’t. 0.4% of the population has cancer. If a person tests positive **twice**, what is the probability they actually have cancer?”*

The AI replied:

Step 1. The test is 95% accurate, so after one positive result the person is 95% likely to have cancer.
Step 2. The two tests are independent, so the probability that both are wrong is $(1 - 0.95)^2 = 0.0025$.
Step 3. Therefore $P(\text{cancer} \mid \text{two positives}) = 1 - 0.0025 = 0.9975 \approx 99.75\%$.

Your tasks:

- B1. Identify **at least two distinct probabilistic errors** in this reasoning. Name each error precisely (which conditional probability is confused with which; which independence statement is misused).

- B2. Write the correct solution assuming the two test results are *conditionally independent given disease status*, using the numbers from class (sensitivity 0.95, specificity 0.96, prevalence 0.004). A clean route: posterior odds = prior odds $\times$ (likelihood ratio)².
- B3. Verify your answer by Monte-Carlo simulation in `HW1.ipynb` (a skeleton cell is provided). Report the estimate and a 95% confidence interval.

Part C — Stress-test the assumption (AI + simulation)

Part B assumed the two test results are conditionally independent given disease status. Suppose instead that, *among healthy people*, a benign condition B (prevalence 2% among the healthy) tends to trigger false positives on **both** tests:

$$P(+ \mid \text{healthy}, B) = 0.9 \text{ per test}, \quad P(+ \mid \text{healthy}, B^c) = p_0,$$

with the two tests conditionally independent given (status, B). Choose p_0 so that the *marginal* false-positive rate stays at 0.04: $0.02 \cdot 0.9 + 0.98 p_0 = 0.04$. Keep the cancer side as in Part B.

- C1. **Before simulating**, discuss with your AI assistant: will $P(\text{cancer} \mid ++)$ go up or down compared to B2, and why? Write down your predicted direction and a one-sentence mechanism.
- C2. Simulate the model (skeleton provided) and estimate $P(\text{cancer} \mid ++)$. Compare with B2 and explain the gap in at most five sentences. *Moral: an independence assumption is a modeling decision, and it can carry most of the answer.*
- C3. (Optional) CP1 #4 used sampling *without* replacement (hypergeometric); CP1 #5 used independent trials (binomial). For a lot of N parts containing exactly $\lceil 0.15N \rceil$ defectives, sample $n = 10$ without replacement and plot the total-variation distance between the hypergeometric pmf and $\text{Binomial}(10, 0.15)$ as N grows. At what N would you call the binomial approximation “safe”?

Part D — Reflection and prompt log (mandatory if AI was used)

Fill the table in `HW1.ipynb` (or reproduce it in your PDF): one row per meaningful AI interaction — where you used it, the prompt, what it returned, and **what you checked or corrected**. Then answer in 2–3 sentences: *where was the AI most useful, and where was it least trustworthy, on this homework?*

Logistics. Solutions to Part A will be posted on PLMS about 10 days after release. Parts B–D have no single “answer key”; selected strong submissions (anonymized) will be discussed in class.